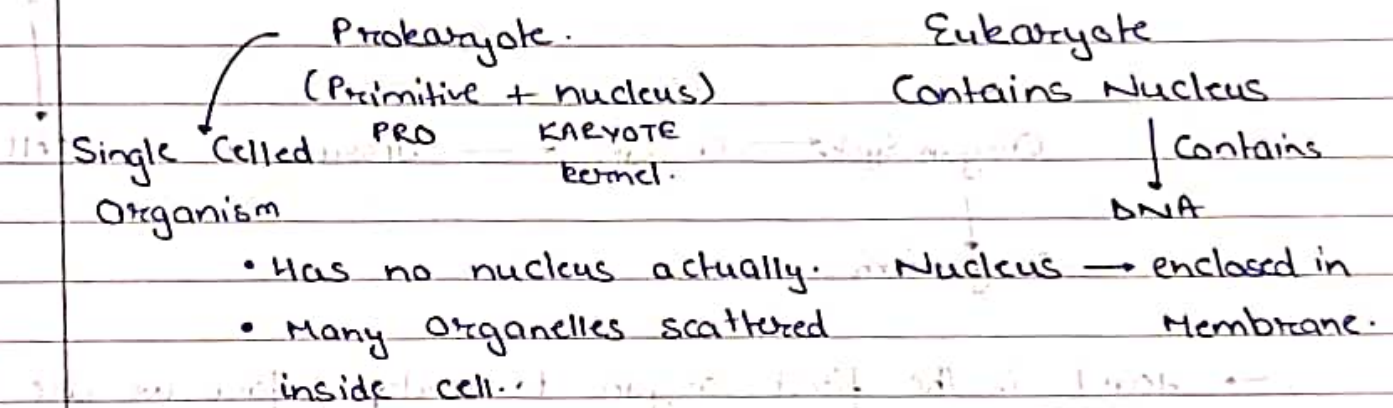


BT-101 (Dr. Rajkumar Thummar)

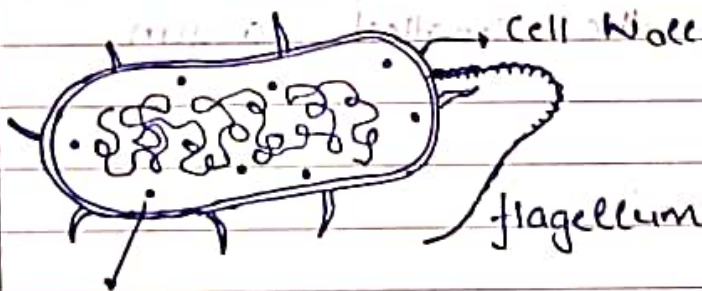
Origin of life: Life emerged 3.8 billion years ago, after 750 million years of Earth formation.
Earth formed $(4.5 - 4.6) \times 10^9$ years ago.

- Cell → Structural and functional unit of living organisms



- length of DNA in one cell = 1.82 m

Humans → HOMO SAPIENS



Ribosomes. Escherichia coli

37.2 trillion cells in the human body.

↳ 10^{12}

6.4 billion base pairs

Half from Mother and Half from father.

Sensory Organs: Eyes, Ears, Nose, Tongue, Skin

__/__/__

→ Including the new discovered organ (Mesentery)

→ 79 Organs in Human Body.

Skin being largest organ present on the human body.

Liver being largest organ inside human body → Regenerative.

→ Respiratory system, Cardiovascular System

Eukaryotes - Homo Sapiens (Human)

Atom → Molecule → Macromolecule → Organelle

↓
Organ System ← Organ ← Tissue ← Cell

↓
Organism

→ Heart is the first organ to develop before we are born and lungs being the last. (Before Birth)

→ Brain keeps on maturing till 25 years of age.

→ Pineal Gland (near the centre of the brain) a groove below the hemispheres is the smallest organ.

• 4 Type of Tissues :-

Epithelial Tissue

Connective Tissue

Muscle Tissue

Nervous Tissue.

(Dies due to change of shape,
enzyme (Haemoglobin) activity decl.)

every haemoglobin molecule
can carry 4 oxygen molecules
(260×10^6 molecules)

120 Days Life. Concave Shaped $\rightarrow$ Contains only Haemoglobin
RBC's and platelets lack DNA (i.e. nucleus). Therefore they do not
undergo cell division

Every Second 12.5 million RBC's are formed using Bone Marrow

(220-200) Different Cell Types.
37.2 Trillion Cells in Human Being

Lysosomes (ph of around 4.5) Killer/Suicide Cells.

Plant Cells do not have centrioles but do possess
cell wall, vacuoles and chloroplasts.

Macromolecules $\rightarrow$ Nucleic Acid, Carbohydrates, Lipids, Proteins.

$\rightarrow$ Human body consists of about 2×10^{25} Molecules, with
99% of it being water

$\rightarrow$ Approximately 7×10^{27} atoms in human body.

$\rightarrow$ 10^{14} atoms in a typical human cell

• Carbon, Hydrogen, Nitrogen, Oxygen, Calcium, Phosphorus
99% of human body.

Mesentery: Largest single continuous stretch of
tissue that supports and position all the digestive
organs in the abdomen.

The organ is responsible for transporting blood and
lymphatic fluid b/w intestine and the rest of the
body.

Mitochondria → It also contains DNA coming entirely from Mother.

Chloroplasts → Also contain DNA material.

→ Single Chromosomes

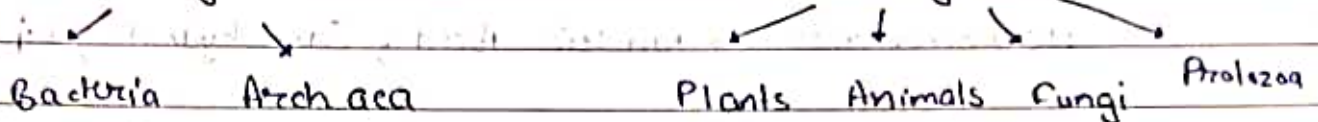
Prokaryotes

Eukaryotes.

- | | |
|---|--|
| 1.) Absence of well defined nucleus. | 1.) Presence of well defined nucleus. |
| 2.) Absence of membrane bound organelles. | 2.) Presence of membrane bound organelles. |
| 3.) Absence of cytoskeleton. | 3.) Presence of cytoskeleton. |
| 4.) Only unicellular | 4.) Both unicellular and multicellular. |
| 5.) Circular DNA. | 5.) Linear DNA, Multiple |
| 6.) 1-10 μm (Size) Organisms | 6.) 10-100 μm (Size) |

Prokaryotes

Eukaryotes



Bacteria reproduce Asexually.

Both reproduction are present.

Prokaryotes

Eukaryotes.

- | | |
|------------------------|------------------------|
| • Presence of Plasmid | • Sexual Reproduction |
| • Asexual Reproduction | • Absence of Plasmids. |

Bacteria) Plasmids have Antibiotic Resistance Genes.

Virus that kills bacteria is called bacteriophages.

Exam
ques
ques

Which of the following proteins are part of cytoskeleton? / /
Which yeast is used in bread mould?

ques.

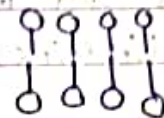
Func of Plasmids?

Virus is non-living.

It comes to life when inside the host.

BT 101 26th March

→ Cell Membrane formed of phospholipids.



→ First RNA came into existence.

→ Scientists believe that once mitochondria and chloroplasts were individual organisms (prokaryotic) (eaten by bigger eukaryotic cells) via phagocytosis

→ 70 s ribosomes and 80 s ribosomes

→ endosymbiosis → Symbiotic Relation in which two organisms coexist and mutually benefit each other via one engulfing other.

→ Basic Molecular Mechanism: Central Dogma of Molecular Biology.

Prokaryotes → DNA

Transcription ↓
RNA

Translation ↓
Protein

Eukaryotes → DNA

Same Mechanism

↓
RNA

↓
Protein

Ques: Is there any protein which converts RNA to DNA in our body? (which)?

- RNA to DNA in virus: Reverse
- Cyanobacteria: First bacteria to learn how to use oxygen.
 - First Photosynthetic Bacteria.

Ques: When did we (human) come into existence? (Timeline)

Apes, chimpanzees almost same genome.
(99.97%)

Ques: Why mitochondria and chloroplasts are considered individual prokaryotic organisms?

- Small Size
- Bc. own DNA material.
- Circular DNA
- Reproduction.

Earth → 4.5 billion years ago.

BT 101

Lynn Margulis

Ques. Which lady proposed the theory of endosymbiosis?

Evolution: It is the change in the inherited characteristics of biological populations over successive generations.

- Bacteria constantly evolves to be resistant to the attack of virus (bacteriophages).

Now, as bacteria's are constantly evolving their protein by genetic mutation. The antibiotics particularly becomes non-functional. Posing problems to doctors.

Procell.

Origin of life on earth is called Protobiogenesis.

Happened around 3.8 billion years ago

Different Theories of Evolution :-

→ Theory of Special Creation :

→ Theory of Spontaneous Generation :

(Life appeared from dirty shirt).

Louis Pasteur, Lazzaro Spallanzani, Francesco Redi → Disapprove this theory.

→ Theory of Chemical Evolution :

→ Theory of Panspermia or Cosmozoic theory :

More than 6.4 billion base pairs

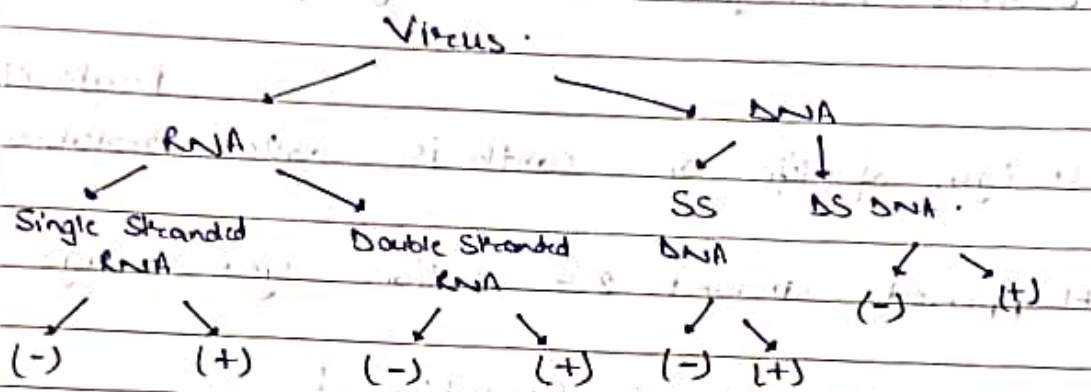
No. of viruses >>> All the species (creatures) present over earth. / /

In gut $\rightarrow$ 150 lakh viruses.

Reproduction $\rightarrow$ Sexual Asexual
(Binary fission).

Virus are called (classified) Non-living as they do not reproduce.

Ribosomes can only accept single stranded RNA.
[We have double stranded DNA]



Ques

What was the genome of corona virus?

Was it DNA or RNA?

Was it the +ve or -ve cells?

CCR5 $\rightarrow$ Gateway for HIV virus.

48 % non-human genome + 40-25 % bacterial genome out of 100 %.

Archaea found over extreme temperature.

Last Universal Common Ancestor → LUCA

__/__/__

→	Gram-negative Bacteria.	Gram Positive Bacteria.
		Thicker peptidoglycan layer.
→	Both IM and OM.	Only inner membrane.
→	Channel is Present.	Channel is Absent.
→	Pore is present.	

Ques. If we apply antibiotics, which one is easier to destroy gram positive or gram negative?

Note :- Phospholipid bilayer is branched in Archaea.